Represent Polygons on the Coordinate Plane

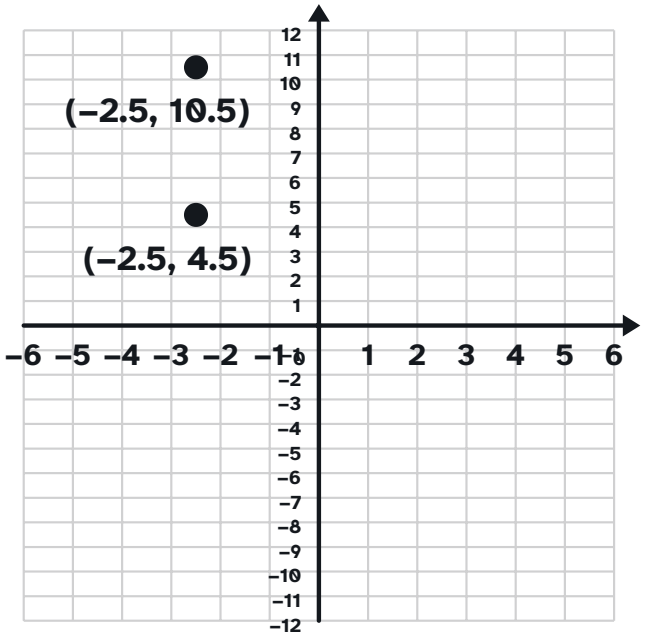
Use coordinates and a scale to find real side lengths and areas.

Name _____ **Date** _____

Tonight's spark

A community garden needs 30,000 square feet. On the plan, one side runs from (–2.5, 10.5) to (–2.5, 4.5). One grid unit is 25 feet. How long is the other side, in feet?

Make a guess first: is the other side more than 100 feet, or less?



Our class model: count units, then convert

Try it

Work it out on paper. Write one answer on each line.

1 **On a plan, 1 unit = 5 feet. A side is 4 units long. How many feet is that?**

A. 9 feet **B.** 20 feet **C.** 4 feet **D.** 45 feet

My answer _____

2 **On a plan, 1 unit = 10 feet. A side is 30 feet long. How many units is that?**

A. 300 units **B.** 40 units **C.** 3 units **D.** 20 units

My answer _____

3 **The points (2, 1) and (2, 7) have the same x. How many units apart are they?**

A. 6 units **B.** 8 units **C.** 7 units **D.** 1 unit

My answer _____

4 **The points (−3, 2) and (−3, −4) have the same x. If 1 unit = 20 feet, how long is the real side?**

A. 40 feet **B.** 6 feet **C.** 26 feet **D.** 120 feet

My answer _____

5 **A plot is 8 units by 5 units. If 1 unit = 10 feet, what is its area in square feet?**

A. 40 square feet **B.** 400 square feet **C.** 260 square feet **D.** 4,000 square feet

My answer _____

Talk about it

How many grid units is the side? What does each unit stand for in real feet?

Home mission

Choose a hallway or a counter. Let one step stand for 2 feet. Ask your student: if the hallway is 7 steps, how many feet is that? Then flip it: a 10-foot wall is how many steps?

If we get stuck, ask

- What does one grid unit stand for here?
- Is this number in grid units or in feet?
- Does this distance cross zero? Then what do we do?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/7-7-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it